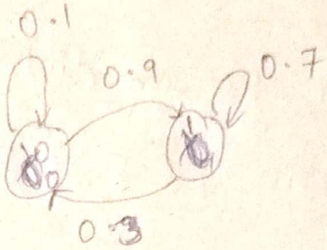
$p^n \rightarrow$ represents n^{th}-step transition probability matrix.

if say $p^n = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$



then $a \rightarrow p(X_n = 0 | X_0 = 0)$

$b \rightarrow p(X_n = 1 | X_0 = 0)$

$c \rightarrow p(X_n = 0 | X_0 = 1)$

$d \rightarrow p(X_n = 1 | X_n = 1)$

$\pi p^n = [\pi_0 \ \pi_1]$ (say)

then $\pi_0 = p(X_n = 0)$

$\pi_1 = p(X_n = 1)$

But while simulating a Markov chain, we ~~get~~ simulate states one by one

<u>Correct answer</u>:-

```
Algo(u1, u2, ---, un):
    states = [ ]
    if u1 < 0.5 : then states.append(0)
                : then states.append(1)
    else
    for i in (1, 2, --, n-1)
        if states.last == 0 : then
            if u_{i+1} < 0.1 then states.append(0)
                             then states.append(1)
            else
        else then
            if u_{i+1} < 0.3 then states.append(0)
                             then states.append(1)
            else
```

P → "one" step probability matrix

$P_{ij} \Rightarrow P(X_k = j \mid X_{k-1} = i) \quad \forall\, k \geq 1$ ($X_0 \rightarrow$ initial state)

→ So once the initial state is ~~st~~ simulated using μ, the ~~probab~~ distribution of the states changes from $[0.5 \quad 0.5]$ to $[0.1 \quad 0.9]$ if $X_0 = 0$ ~~is the initial state~~

$[0.3 \quad 0.7]$ if $X_0 = 1$

which is different from

$$\pi P = [0.5 \quad 0.5] \begin{bmatrix} 0.1 & 0.9 \\ 0.3 & 0.7 \end{bmatrix} = [0.2 \quad 0.8]$$

→ at any time t, the distribution of states depends on the previous state but $\pi P^{(t-1)}$ doesn't depend on the previous state. It represents the probability of X_t marginalized (which is not conditioned on anything)

→ Using $\pi P^{(n)}$ will generate a simulation which doesn't represent a markov chain. By using $\pi P^{(n)}$ we're ~~you~~ removing the dependence of the future state of a markov chain on it's ~~previ~~ present state which is not even a markov chain anymore.

A simple contradiction:-

$\pi P^{(n)}$ method	Correct Method
$p(X_0=0, X_1=1) = p(X_0=0)\,\underbrace{p(X_1=1/X_0=0)}_{[\pi P]_{12}}$ $= 0.5 \times 0.8 = 0.4$ $= p(X_0=0)\cdot p(X_1=1)$ ↓ shows that X_0 & X_1 are independent ↓ Wrong	$p(X_0=0, X_1=1)$ $= \underbrace{p(X_0=0)}_{0.5}\cdot \underbrace{p(X_1=1/X_0=0)}_{0.9}$ $= 0.45$ $\neq p(X_0=0)\cdot p(X_1=1)$

according to the $\pi P^{(n)}$

$p(X_0=0)\cdot \underbrace{p(X_1=1/X_0=0)}$

$\pi P = [\pi P_{11} \quad \pi P_{12}]$
$\qquad\quad 0.2 \qquad 0.8$

$\pi P^{(n)}$ methods substitutes this with $[\pi P]_{12} = P(X_1=1)$

$\Rightarrow p(X_0=0)\cdot p(X_1=1)$

(It says $p(X_0=0, X_1=1) = p(X_0=0)\cdot P(X_1=1)$)

$\Rightarrow X_0$ & X_1 are independent $\Rightarrow$ Not a markov chain

$\Rightarrow$ Contradiction